

Feature Enhancement: Smart Insights Section for ElectricityLoadCalculator.com

Context

The website already provides:

- Appliance-based load calculation
- Total Load (kW/W)
- Daily Consumption (kWh/day)
- Monthly Consumption (kWh/month)
- Estimated Monthly Bill
- Top Consumer Analysis
- PDF Export

The current UI uses a dark theme with yellow/amber accents and a clean mobile-first card layout.

Do NOT redesign the existing summary report.

The goal is to enhance the report page by introducing a new "Smart Insights" section directly below the existing Load Summary and Top Consumer sections.

Objective

Transform the calculator from a "result calculator" into an "electrical advisor."

Instead of only showing numbers, provide actionable recommendations based on the calculated load.

UI Placement

Insert a new section after:

- Total Load
- Daily Consumption
- Monthly Consumption
- Estimated Bill
- Top Consumer

New Section Title:

Smart Insights

Use the same design language as the existing site:

- Dark cards
 - Rounded corners
 - Yellow accent color
 - Consistent spacing
 - Mobile-first responsive design
-

Smart Insight Cards

Create four insight cards.

Each card should contain:

- Icon
 - Recommendation value
 - Short explanation
 - View Details CTA
-

Card 1: MCB Recommendation

Display:

Recommended MCB

Example:

16A C-Curve

Explanation:

Based on your calculated electrical load and a standard safety margin.

Button:

View Details →

Card 2: Cable Recommendation

Display:

Recommended Cable Size

Example:

2.5 sq mm Copper

Explanation:

Suitable for the recommended current carrying capacity.

Button:

[View Details →](#)

Card 3: Inverter Recommendation

Display:

Recommended Inverter

Example:

3 kVA

Explanation:

Based on the connected load.

Button:

[View Details →](#)

Card 4: Solar Recommendation

Display:

Recommended Solar System

Example:

5 kW Solar Plant

Explanation:

Sized according to estimated monthly consumption.

Button:

[View Details →](#)

Insight Calculation Logic

Create a reusable Insight Engine.

Input:

totalLoadKW monthlyUnits dailyUnits

Output:

{ recommendedMCB, recommendedCable, recommendedInverter, recommendedSolar }

MCB Recommendation Logic

Single phase assumption:

$\text{Current} = (\text{totalLoadKW} \times 1000) / 230$

Apply 25% safety margin.

Select nearest standard rating:

[6,10,16,20,25,32,40,63,80,100]

Examples:

0-1 kW → 6A

1-2 kW → 10A

2-3 kW → 16A

3-5 kW → 25A

5-7 kW → 32A

7-10 kW → 40A

10-15 kW → 63A

Return:

"16A C-Curve"

Cable Recommendation Logic

Suggested Copper Cable:

6A → 1.0 sq mm

10A → 1.5 sq mm

16A → 2.5 sq mm

25A → 4 sq mm

32A → 6 sq mm

40A → 10 sq mm

63A → 16 sq mm

Return:

"2.5 sq mm Copper"

Inverter Recommendation Logic

Recommended inverter size:

$\text{inverterKVA} = \text{totalLoadKW} \times 1.2$

Round up to nearest standard:

1 kVA

2 kVA

3 kVA

5 kVA

7.5 kVA

10 kVA

15 kVA

Return:

"3 kVA"

Solar Recommendation Logic

Estimate solar size:

$\text{solarKW} = \text{monthlyUnits} / 120$

Round to nearest 0.5 kW.

Example:

753 units/month

$753 \div 120$

≈ 6.3

Return:

"6.5 kW Solar System"

View Details Navigation

Each card must support navigation.

Routes:

/insights/mcb

/insights/cable

/insights/inverter

/insights/solar

For now:

Create placeholder pages.

Pass calculated values through route state, query params, or global store.

Do not implement advanced calculators yet.

Only prepare navigation architecture.

PDF Export Enhancement

Add a new page section inside the generated PDF.

Title:

SMART INSIGHTS

Include:

Recommended MCB

Recommended Cable

Recommended Inverter

Recommended Solar

Format should match the existing report style.

Do not modify existing sections.

Append Smart Insights after the summary and before the appliance breakdown.

Technical Requirements

- Reuse existing calculation results.
 - No duplicate calculations.
 - No backend changes.
 - Fully client-side.
 - Mobile-first responsive design.
 - Maintain current dark/yellow visual identity.
 - Keep performance impact minimal.
 - Follow existing component structure and design system.
-

Success Criteria

After calculation, users should immediately see:

 Recommended MCB

 Recommended Cable

 Recommended Inverter

 Recommended Solar

without leaving the report page.

The experience should feel like a professional electrical assessment rather than a simple load calculator.